

Solution Requirements

Date	15 March 2026
Team ID	xxxxxx
Project Name	SmartLender-ApplicantCredibility Prediction for Loan Approval
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can securely register and log in using email or mobile number with OTP/password verification.	High
2	Authorization levels	Customers can apply for loans and view recommendations, while administrators can manage user, loan products & approvals	High
3	External interfaces	Integrate with bank APIs, credit score services, and document verification systems to retrieve loan and customer information.	High

4	Transactions processing	The system securely processes loan applications, verifies customer details, and updates application status in real time.	High
5	Reporting	The system generates reports on loan applications, approvals, rejections, customer details, and loan recommendations for administrator	Medium
6	Business rules, etc.	Loan recommendations are based on customer eligibility, income, credit score, repayment capacity, and bank policies.	High
7	Compliance to laws or regulations	The system protects customer data, follows banking regulations, and complies with data privacy and security standards.	High
8	<i>Other</i>	The system provides AI-based loan recommendations, sends application status notifications, and offers customer support through a chatbot.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should process loan requests and chatbot responses quickly with minimal delay.	Page loads in less than 2 seconds and chatbot responds within 3 seconds.
2	Scalability	The system should support increasing numbers of users and loan applications without performance degradation	Supports up to 10,000 concurrent users.
3	Security & Data Privacy	User personal and financial data must be securely stored and transmitted using encryption and secure authentication.	AES-256 encryption, HTTPS, and secure user authentication.
4	Reliability & Availability	The application should be available at all times with minimal downtime and reliable loan processing.	99.9% uptime per month.
5	Usability & Accessibility	The system should have a simple, user-friendly interface that is easy to access from desktop and mobile device.	Easy navigation with mobile friendly design and accessibility compliance.
6	Other	The system should be easy to maintain, update, and integrate with future banking or financial services.	Supports regular Updates, bug fixes, and API integration without Affecting users.